

Chin-Cheng Chan

Los Angeles, CA | chincheng@usc.edu | [Google Scholar](#) | [Personal Website](#)

RESEARCH SUMMARY

My research applies signal processing theory and mathematics to **develop rigorous, high-quality, efficient, and trustworthy image reconstruction algorithms** for ill-posed inverse problems in computational imaging. Representative works of mine include (1) a novel framework for designing reconstruction methods that balance quality, efficiency, and transparency, (2) a computationally efficient image model that addresses a 25-year-old computational challenge in non-Cartesian Fourier imaging, and (3) a framework based on perturbation analysis to characterize the resolution of arbitrary reconstruction algorithms. Many of the algorithms I developed have a **broad impact on both classical and modern machine-learning reconstruction methods** and are **applicable across multiple imaging modalities**.

EDUCATION

University of Southern California, Los Angeles, CA

Aug. 2019 — Present

- Ph.D. in Electrical and Computer Engineering, expected May 2026.
- Research areas: image reconstruction, computational imaging, and MRI.

National Taiwan University, Taipei, Taiwan

Sep. 2014 — Jun. 2018

- B.Sc. in Electrical Engineering.
- Research areas: image processing for camera systems and computer vision.

RESEARCH EXPERIENCE

Biomedical Imaging Group, University of Southern California

Los Angeles, CA

Graduate Research Assistant (Advisor: Prof. Justin P. Haldar)

Aug. 2019 — Present

- A Novel Hilbert-Space Reconstruction Framework for Computational Imaging
 - Developed a novel framework for designing high-quality, efficient, and trustworthy image reconstruction methods using reproducing-kernel Hilbert space theory.
 - Addressed a long-standing trade-off between accuracy, computational efficiency, and transparency in the modern reconstruction landscape.
 - Achieved 6× speedup in MRI reconstruction without loss of image quality.
- Computationally Efficient Image Model for Non-Cartesian Fourier Imaging
 - Proposed a novel image model for non-Cartesian Fourier imaging.
 - Addressed a long-standing computational efficiency challenge in the conventional model used over the past 25 years.
 - Achieved 2×–4× faster reconstruction compared to the conventional model.
- Resolution Analysis Framework for Modern Reconstruction Algorithms
 - Developed a framework based on perturbation analysis to characterize the resolution of arbitrary reconstruction algorithms.
 - Tackled the emerging challenge of evaluating resolution in reconstructions generated by data-driven and advanced nonlinear methods.
 - Showed that commonly used image-quality metrics may not reliably reflect image resolution.
- Novel Beamforming Technique for Cartesian MRI
 - Proposed an enhanced beamforming technique tailored for Cartesian MRI acquisitions.
 - Mitigated the fundamental trade-off between scan time and reconstruction time in MRI.
 - Enabled 50× faster image reconstruction for highly accelerated real-time MRI compared to conventional methods.

Multimedia Processing and Communications Lab, National Taiwan University

Taipei, Taiwan

Undergraduate Research Assistant (Advisor: Homer Chen)

Aug. 2015 — Jun. 2019

- Deep Learning for Optical Coherence Tomography (OCT) Image Analysis
 - Developed a deep learning method for delineating the dermis-epidermis junction and reduced segmentation error by 40%.

- Improving the Reliability of Phase-Detection Autofocus
 - Proposed autofocus techniques using statistics, reinforcement learning, and deep learning, achieving $1.7\times$ speedup.
- Blood Vessel Detection for Optical Coherence Tomography
 - Proposed a short-time robust PCA algorithm that improved the classification performance by 20%.

PEER-REVIEWED PUBLICATIONS

Preprint/Under Review

1. C.-C. Chan, J. P. Haldar, “A new k-space model for non-Cartesian Fourier imaging”, *arXiv:2505.05647*, 2025. (Under review). [[Link](#)]

Journal publications

1. R. Lobos, C.-C. Chan, J. P. Haldar, “New theory and faster computations for subspace-based sensitivity map estimation in multichannel MRI”, *IEEE Trans. Med. Imag.*, vol. 43, pp. 1–10, 2023. [[Link](#)]
2. S.-T. Tsai, C.-H. Liu, C.-C. Chan, Y.-H. Li, S.-L. Huang, H. H. Chen, “H&E-like staining of OCT images of human skin via generative adversarial network”, *Appl. Phys. Lett.*, vol. 121, pp. 134102, 2022. [[Link](#)]
3. C.-C. Chan, J. P. Haldar, “Local perturbation responses and checkerboard test: A tool for characterizing advanced nonlinear algorithms for inverse problems in MR”, *Magn. Reson. Med.*, vol. 86, pp. 1873–1887, 2021. [[Link](#)]
4. C.-J. Ho, M. Calderon-Delgado, C.-C. Chan, M.-Y. Lin, J.-W. Tjiu, S.-L. Huang, H. H. Chen, “Detecting mouse squamous cell carcinoma from submicron full-field optical coherence tomography images by deep learning”, *J. Biophotonics*, vol. 14, 2021. [[Link](#)]
5. C.-J. Ho, C.-C. Chan, H. H. Chen, “AF-Net: A convolutional neural network approach to phase detection autofocus”, *IEEE Trans. Image Process.*, vol. 29, pp. 6386–6395, 2020. [[Link](#)]
6. C.-C. Chan, H. H. Chen, “Modeling phase shift data of phase-detection autofocus by skew-normal distribution”, *J. Electron. Imaging*, vol. 28, 2019. [[Link](#)]
7. P.-H. Lee, C.-C. Chan, S.-L. Huang, A. Chen, H. H. Chen, “Extracting blood vessels from full-field OCT data of human skin by short-time RPCA”, *IEEE Trans. Med. Imag.*, vol. 37, pp. 1899–1909, 2018. [[Link](#)]

Conference publications

1. C.-C. Chan, J. P. Haldar, “Constrained MRI using weighted Hilbert spaces: Fast scan-specific reconstruction with transparent assumptions”, *ISMRM Workshop on Data Samp. & Image Recon.*, 2026. In Press. (Abstract) [[Link](#)]
2. C.-C. Chan, J. P. Haldar, “A novel k-space model for non-Cartesian reconstruction”, in *Proc. Int. Soc. Magn. Reson. Med.*, 2025, p. 1366. (**Magna Cum Laude Award**). (Abstract) [[Link](#)]
3. C.-C. Chan, D. Kara, D. Kwon, E. Roselli, C. Nguyen, J. P. Haldar, “ROVir enables substantially easier real-time imaging of small regions of interest”, in *Proc. Int. Soc. Magn. Reson. Med.*, 2025, p. 2622. (Abstract) [[Link](#)]
4. C.-C. Chan, J. P. Haldar, “Rethinking model-based non-Cartesian Fourier imaging: A new k-space model”, in *Proc. IEEE Int. Symp. Biomed. Imag.*, 2025. [[Link](#)]
5. C.-C. Chan, C. Nguyen, J. P. Haldar, “Improved region-optimized virtual coils for Cartesian acquisition geometries”, in *Proc. Int. Soc. Magn. Reson. Med.*, 2024, p. 1905. (Abstract) [[Link](#)]
6. C.-C. Chan, J. P. Haldar, “Measuring spatiotemporal resolution in real-time MRI”, in *Proc. Int. Soc. Magn. Reson. Med.*, 2024, p. 1875. (Abstract) [[Link](#)]
7. C.-C. Chan, J. Wang, T. Nadeem, J. P. Haldar, “On reference-based image quality assessment in medical image reconstruction: Potential pitfalls and possible solutions”, in *Proc. Asilomar*, 2023, pp. 36–39. (Invited paper) [[Link](#)]
8. R. Lobos, C.-C. Chan, J. P. Haldar, “New theory and faster computations for subspace-based sensitivity map estimation”, in *Proc. Int. Soc. Magn. Reson. Med.*, 2023, p. 4625. (Abstract) [[Link](#)]
9. C.-C. Chan, J. P. Haldar, “Local perturbation responses: A tool for understanding the characteristics of advanced nonlinear MR reconstruction algorithms”, in *Proc. Int. Soc. Magn. Reson. Med.*, 2020, p. 684. (Abstract) [[Link](#)]
10. C.-C. Chan, H. H. Chen, “Autofocus by deep reinforcement learning”, in *Proc. Electron. Imaging*, 2019, pp. 577–581. [[Link](#)]
11. C.-C. Chan, H. H. Chen, “Improving the reliability of phase detection autofocus”, in *Proc. Electron. Imaging*, 2018, pp. 1–5. [[Link](#)]
12. C.-C. Chan, S.-K. Huang, H. H. Chen, “Enhancement of phase detection for autofocus”, in *Proc. IEEE Int. Conf. Image Process.*, 2017, pp. 41–45. [[Link](#)]

13. P.-H. Lee, **C.-C. Chan**, S.-L. Huang, A. Chen, H. H. Chen, “Blood vessel extraction from OCT data by short-time RPCA”, in *Proc. IEEE Int. Conf. Image Process.*, 2016, pp. 394-398. [[Link](#)]

AWARDS

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|---|-------------------------|
| • Ming Hsieh Institute Scholar | Oct. 2025 |
| • Magna Cum Laude Award at ISMRM 2025. | May 2025 |
| • Educational stipends for ISMRM 2024 and 2025. | Jan. 2024 and Jan. 2025 |

TEACHING AND MENTORING EXPERIENCE

Peer Mentor, University of Southern California

Los Angeles, CA

- Mentored a junior PhD lab member on research projects.

Jul. 2025 — Present

Teaching Assistant, University of Southern California

Los Angeles, CA

- EE-483: Graduate-level Digital Signal Processing.

Fall 2022, Fall 2023

- EE-503: Graduate-level Engineering Probability.

Fall 2021, Spring 2022, Spring 2024

- PHYS-152L: Lab section for Physics II.

Fall 2024